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Author not stated, Publisher not stated.

- Published: date not stated
- Original: <<https://trouge.net/publication/domclob-sp-2023/>>
- Also published at: <https://trouge.net/papers/domclob_sp23.pdf>
- Preserved from: https://trouge.net/papers/domclob_sp23.pdf (live) on 2026-08-10
- Licence: unknown

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It's (DOM) Clobbering Time: Attack Techniques, Prevalence, and Defenses

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Abstract—DOM Clobbering is a type of code-less injection markups. The research community has only recently started attack where attackers insert a piece of non-script, seemingly studying the security of these interactions, mainly focusing benign HTML markup into a webpage and transform it on small code fragments called script gadgets [12] that to executable code by exploiting the unforeseen interactions react to injected HTML markups and transform it into code. between JavaScript code and the runtime environment. The Unfortunately, script gadgets are only the tip of the iceberg, attack techniques, browser behaviours, and vulnerable code and other complex interactions exist that attackers can abuse patterns that enable DOM Clobbering has not been studied to hijack the program execution, which, to date, are largely yet, and in this paper, we undertake one of the first evaluations unexplored. of the state of DOM Clobbering on the Web platform. Starting DOM Clobbering is a vulnerability that originates from with a comprehensive survey of existing literature and dynamic a naming collision between JavaScript variables and named analysis of 19 different mobile and desktop browsers, we sys- HTML markups, where browsers replace pre-existing content tematize DOM Clobbering attacks, uncovering 31.4K distinct of an undefined variable with an HTML element when the markups that use five different techniques to unexpectedly variable name and the element's name (or id) attribute overwrite JavaScript variables in at least one browser. Then, we match. Developers unaware of such behavior may use the use our

systematization to identify and characterize program content of undefined variables for sensitive operations, such instructions that can be overwritten by DOM Clobbering, and as URLs for fetching remote content, and attackers can use it to present TheThing, an automated system that detects exploit it by injecting markups with colliding names. DOM Clobbering vulnerabilities have been known for over a decade, with the first instance identified in 2010 [13] where instantiate TheThing on the top of the Tranco top 5K sites, an iframe named self allowed attackers to overwrite quantifying the prevalence and impact of DOM Clobbering the top window location of webpages containing frame- in the wild. Our evaluation uncovers that DOM Clobbering busting code, i.e., assignments such as top.location vulnerabilities are ubiquitous, with a total of 9,467 vulnerable = self.location. Since then, security researchers have data flows across 491 affected sites, making it possible to mount identified new, more subtle attack variants, combining pairs of arbitrary code execution, open redirections, or client-side re- HTML tags (e.g., [14, 15]) or browser-specific markups and quest forgery attacks also against popular websites such as attributes (e.g., [16–19]), and clobbering not only variables, Fandom, Trello, Vimeo, TripAdvisor, WikiBooks and GitHub, but also deep object properties (e.g., [20–22]), nested window that were not exploitable through the traditional attack vectors. proxies (e.g., [21, 23]) and loops (e.g., [21]). When looking Finally, in this paper, we also evaluate the robustness of at the possible combinations of tags, attributes, code features, the existing countermeasures, such as HTML sanitizers and and runtime behaviors, prior works have merely scratched Content Security Policy, against DOM Clobbering. the attack surface, and, to date, we still miss a systematic and comprehensive exploration of this threat. Index Terms—DOM Clobbering, Attack Techniques, Preva- Recently, DOM Clobbering vulnerabilities in Gmail [22] lence, Defenses and Google Analytics [24, 25] revamped new discussions about defenses, such as proposing to switch off named

1. Introduction property accesses for DOM elements at the browser level (see, Arbitrary client-side JavaScript execution has been one of e.g., [25–27]), which has been dismissed since, according the major threats against web applications since the early to Google Chrome telemetry data, about 10.5% of the days, traditionally achieved by injecting JavaScript code into pages in 2021 use named property accesses to implement vulnerable pages, e.g., Cross-Site Scripting (XSS) attacks [1– functionalities that could otherwise break [28]. To date, the 11]. However, over the past 20 years, the growth of Web burden of protecting from DOM Clobbering attacks is solely technology has introduced unforeseen interactions between on developers’ hands, who can use existing countermeasures JavaScript programs and the execution environment that such as HTML sanitizers tailored to protect against DOM can result in execution of arbitrary code without injecting Clobbering, e.g., DOMPurify [7], or mitigate the risk of JavaScript but only by injecting seemingly harmless HTML code execution via Content Security Policy (CSP) [29– 31].

1 Unfortunately, DOMPurify protects only from specific DOM Listing 1: Example of DOM Clobbering vulnerability where named proper- ties overshadow JavaScript variables. Clobbering cases, whereas CSP cannot prevent the execution 1 var s =

document.createElement('script'); of already-present code that reacts to markup injections, 2 let config = window.globalConfig || {href: 'script.js'}; suggesting that existing countermeasures may be incomplete 3 s.src = config.href; or even insufficient. As a last resort, developers can develop 4 document.body.appendChild(s); their own defenses, requiring a deep understanding of the main threat and its variants, which, unfortunately, may not be the case. For example, as witnessed by recent DOM Listing 2: Example of DOM Clobbering vulnerability where named proper- Clobbering vulnerabilities discovered in HTML sanitizers, ties overshadow native DOM APIs. e.g., DOMPurify [17] and HTML Janitor [20], developers 1 var s = document.createElement('script'); 2 let b = document.documentElement.getAttribute('baseURI'); may still be largely unaware of the risk posed by DOM 3 s.src = b + '/script.js'; Clobbering vulnerabilities. 4 document.body.appendChild(s);

In this paper, we take a step back and take a deep look at DOM clobbering, with, to the best of our knowledge, the more breakage, i.e., 2,561 websites, than benefits, i.e., 491 first systematic and comprehensive study on this neglected vulnerable websites, with a cost-benefit ratio of 5.2:1 web- vulnerability class, covering three main aspects: a systematic sites. In the absence of a browser-level fix, developers need to exploration of the attack surface, a measurement of affected be particularly careful when choosing a countermeasure, as and vulnerable websites, and a review and evaluation of they balance protection with usability. For example, 55% of defenses. Starting from a comprehensive survey of prior the most popular HTML sanitizers across the five most used DOM Clobbering vulnerabilities, we systematically generate web languages are vulnerable to at least one of the 31.4K candidate DOM Clobbering markups, and automatically test clobbering markups by default. The remaining 45% sanitizers desktop and mobile browsers against them, covering all remove named properties, i.e., id and name attributes, which known HTML tags and attributes—including custom ones—and may interfere with the DOM manipulation operations. Also, markup relationships. Then, we propose TheThing, a DOM our results show that CSP is insufficient because 85% of the Clobbering detection tool that combines hybrid program discovered vulnerabilities can cause code execution without analysis, i.e., [32], for the discovery of potentially-vulnerable manipulating the src attribute. Finally, our results show that data flows, with forced execution, i.e., [33], for the automated developers can fix vulnerabilities at the code level, and we vulnerability verification, leveraging the generated DOM identify eight distinct vulnerable code patterns to avoid and Clobbering markups. We instantiate TheThing against the propose four secure patterns to fix them. Tranco top 5K websites to quantify the prevalence and impact of DOM Clobbering vulnerabilities, processing, in Contributions. To summarize, this paper makes the follow- total, over 24.6B lines of JavaScript code across 18.3M ing contributions:

- We conduct the first comprehensive and systematic scripts and 205.6K webpages. Finally, we identify, review, and evaluate defenses, covering existing countermeasures and study of DOM Clobbering, covering vulnerability, attack secure code patterns. In particular, we first precisely measure techniques, detection, prevalence, impact, and defenses.
- We propose a systematic technique to identify DOM the cost-benefit trade-off of browser-level countermeasures and thoroughly test HTML sanitizers. Then, we review the Clobbering markups and test browsers automatically, vulnerable code discovered by TheThing, identify common identifying 148 previously-

unknown ones, 30,803 new developer mistakes, and distill a list of secure coding patterns. variants, and 114 new browser APIs that can be clobbered in at least one browser. Our results show that the attack surface of DOM Clobbering is large, with only 481 out of 31,432 generated DOM Clobbering markups are currently known, DOM Clobbering that uncovered 9,467 DOM Clobbering and the remainings are either previously-unknown instances bearing vulnerabilities, affecting 9.8% of the Tranco (148) or variants of known cases (30,803). When grouping top 5K sites, of which 44 that we manually confirmed markups by browser behaviors, we observe ten different to be exploitable, including popular sites like GitHub, behavioral groups, showing that while most of the attacks Fandom, Vimeo, Trello, TripAdvisor, and AliExpress. We evaluate the robustness of 29 client-side and server-side HTML sanitizers and CSP, showing that 55% new native browser APIs that these markups clobber in of sanitizers are vulnerable and 85% of the DOM at least one browser, including security-sensitive APIs like Clobbering vulnerabilities cannot be mitigated by CSP. We review existing countermeasures, analyze common cache storage [34] and trusted types [35]. Second, DOM Clobbering vulnerabilities are quite widespread, affecting mistakes of the 491 vulnerable sites, and distill a list 9.8% of the top 5K websites, including popular sites like of recommendations and secure coding patterns. GitHub, Fandom, Trello, Vimeo, TripAdvisor, WikiBooks and AliExpress, leading to severe consequences such as

1. Background

arbitrary code execution, client-side CSRF [32], and open Before presenting our study, we first dissect and introduce redirections [36, 37]. Third, when looking at the browser- the DOM Clobbering vulnerability in §2.1, and then, we level defenses, disabling named property accesses can cause present the threat model of this work in §2.2.

2.2.1. DOM Clobbering Vulnerability web attacker [46, 47] can achieve that, e.g., adding a preview of a post to the client-side webpage by leveraging the URL DOM Clobbering vulnerabilities originate from a naming parameters. Another example is the case where the attacker collision between JavaScript variables and named HTML can implant a persistent DOM Clobbering payload in the markups, i.e., markups with an id or name attribute [15, target webpage, which can lie dormant, and exploited later 38, 39]. When an undefined variable [40, 41] and an HTML on to attack a victim, e.g., adding persistent comments in the markup have the same name, the browser replaces the pre-UI through Gmail's dynamic email feature [48] which allows existing content of the variable with the DOM object mirror- including HTML content [22], or user-generated Markdown in the markup type. Listing 1 shows a snippet of vulnerable descriptions in code repositories that are turned into HTML code, which loads a script whose URL is stored in a global content [49, 50]. Finally, a more powerful web attacker configuration object, i.e., `window.globalConfig`. In (e.g., [5, 12]) who is aware of a markup injection vulnerability more details, the code first creates a script tag (line 1), and can manipulate the DOM tree. then, it retrieves the global configuration object and stores it in a local variable `config` (line 2). If the configuration 3. Problem Statement object does not exist, it uses a minimal default

configuration, i.e., {href: 'script.js'} (line 2). Then, the program sets the src attribute of the newly created script tag to (RQ1) DOM Clobbering Attack Techniques. When looking at the evolution of DOM Clobbering attack markups, we observe a consistent complexity growth, starting from a single HTML element [13] that can overwrite a variable, evolving line 2 because attackers can control the value of with pairs of HTML tags [14, 15] that clobber properties of window.globalConfig, and ultimately, pick the objects (2013-15), and then advancing into a wide variety script src value of their choosing by injecting an of browser-specific combinations of different HTML tags HTML tag with id="globalConfig", e.g., <a and attributes that can not only overwrite variables, but id="globalConfig" href="malicious.js">. also native DOM objects (2015-18) [16-19], nested object When parsing such a markup code, the browser maps properties, and loop elements (2018-22) [21-23]. Despite the the anchor tag element to the window.globalConfig growth of markups' complexity, the exploration of the attack property as mandated by the named property access surface has not been conducted systematically, and to date, rule of the HTML specifications (see [42-44]). The many of the possible combinations of tags, attributes, markup escalation to arbitrary code execution happens in line 3, relationships and possible JavaScript object manipulations when the code reads the href property of the object are not considered. As a first research question, we intend to window.globalConfig, which no longer contains the fill this gap and exhaustively explore such an attack surface object with the global configuration but it contains the by generating clobbering markups and testing modern mobile attacker-controlled anchor tag whose href property value and desktop browsers automatically. is malicious.js. (RQ2) Detection, Prevalence and Impact. While the ex- Attackers can abuse named property accesses in other istence of DOM Clobbering is known for more than a ways, where instead of overwriting variables by HTML nodes, decade [13, 14], we still do not have a measurement about they can overshadow browser APIs. Listing 2 illustrates an ex- the prevalence, impact, and code patterns of this vulnerability. ample of such an attack. Similarly to Listing 1, this code also In this paper, we intend to quantify the prevalence of DOM dynamically creates and loads a script. Instead of fetching Clobbering in the wild, identify vulnerable behaviours, and the URL from a global configuration object, the code intends examine their impact to shed some light on possible causes to use the baseURI attribute of the main HTML tag via the and factors hampering web applications' security. document.documentElement API (line 2). An attacker (RQ3) Defenses and Effectiveness. As a final question, we can manipulate the content of src in line 3 by shadow- look at the defenses, their effectiveness, and cost-benefit, ing the native property document.documentElement leveraging the data generated and collected from the previ- using an attacker-injected node in the DOM tree [45], ous answers, i.e., DOM Clobbering markups, vulnerability e.g., a form element with name="documentElement" prevalence, and developer mistakes. In particular, we intend and the custom property baseURI="malicious.js". to re-evaluate the cost-benefit trade-off resulting from dis- When parsing the form tag, the browser maps the prop- abling named property accesses in browsers and thoroughly erty document.documentElement to the JavaScript assess existing solutions such as HTML sanitization [7], object representing the

form tag (an instance of the Content-Security Policy (CSP) [22, 29], and freezing object HTMLFormElement class) which has a function called properties [51] against DOM Clobbering. Finally, we intend to getAttribute which returns the value of the attribute to review developers' mistakes and identify vulnerable and baseURI, i.e., the string malicious.js. secure coding patterns that can fix those issues.

2.2. Threat Model

4. Attack Techniques

In a DOM Clobbering attack, the attacker needs to insert an HTML payload into a target, vulnerable webpage. At the different ways DOM Clobbering markups can manipulate

JavaScript variables, object properties, and native APIs. the target is clobbered. Then, we instantiate each browser Before presenting our findings (§4.2), we describe the methodology we followed to answer this RQ (§4.1). used BrowserStack [59] to programmatically control browser versions, names, and their execution life-cycle in a fully automatic fashion. We evaluated (the latest versions of) all mobile and desktop browsers available in BrowserStack (i.e., review existing works on DOM Clobbering attacks, looking at 16 browsers), and additionally tested the Tor Browser for the various techniques to generate markups and at the sake of completeness. Finally, for Safari, we considered browser specifications causing the overrides. Then, we apply three different versions that correspond to the three recent macOS operating systems as Safari cannot be upgraded and thoroughly test browsers. standalone [60]. In total, we evaluated 19 browsers. Overall, our generation algorithm produced 3,906,136 candidate test markups, of which 34,648 are for targets As the first step, we systematically reviewed the existing 'x', i.e., variables or native APIs, and the rest are for object literature on DOM Clobbering attack markups, i.e., the academic properties 'x.y' and 'x.x'. When testing variables, we replace academic literature [7, 12, 13, 26], HackerOne vulnerability re- the target 'x' with the variable name generating in total ports [52], the CVE database [53], Bugzilla bug reports [16], 34,648 test cases for variables. When testing native DOM and non-academic resources (see, i.e., [14, 15, 17, 21- APIs, we replace the target 'x' with the API function or [23, 39, 54, 55]). Then, for each discovered DOM Clobbering property name (e.g., the cookie property of document), instance, we extracted the HTML tags, attributes, the clobber- obtaining 34,648 test cases per API function. As of October 2021, the total number of DOM API objects is 581 [61], the object type of the clobbered target (e.g., HTMLElement of which 347 are window APIs (i.e., 291 properties and or WindowProxy), and tags relation (i.e., child, srcdoc, 56 methods) [62], and 234 APIs are for the document or sibling). Then, we looked for the corresponding browser object (i.e., 178 properties and 56 methods) [63]. In total, specification rules that explain the reason why the clobbering we generated 20,130,488 test cases for native APIs. instance works. When the rule defines other variants of the clobbering instance, we add them to the list of the 4.2. Results instances. Accordingly, we reviewed the HTML and DOM This section presents the results of our literature review specifications [56, 57], and GitHub issues in the specifications' and browser testing. repositories, i.e., W3C permissions policy [27], WICG document policy [19, 25], and WHATWG HTML and DOM

4.2.1. Systematization of Known Instances

standard repositories

[18, 58]. Finally, we group instances together based on their similarity, i.e., tags, attributes, target, Table 1 summarizes the DOM Clobbering markups. Our and the type of the value it refers to. Table 1 shows the review identified 481 DOM Clobbering instances that we result of our systematization. grouped into 13 classes based on their structural similarity. Each instance shows how a specific HTML markup (e.g., 4.1.2. Markup Generation and Browser Testing) can clobber a specific target, i.e., variable Starting from our systematization, we derived a list of (e.g., x) or object property (e.g., window.x), and replaced rules for generating DOM Clobbering markups, covering it with a JavaScript object (e.g., x is shadowed by an all HTML tags, attributes, tags' relations, and attack targets HTMLAnchorElement). For each class, the table shows (i.e., a variable, an object property, or a native browser API). the clobbered target, the HTML code that can overwrite it, First, we generated candidate HTML markups for a target and the object type stored in the target. Also, the review 'x' using all the 142 valid HTML tags, including a custom of the HTML and DOM specifications resulted in the tag (e.g., mytag), and all the 244 valid HTML attributes, identification of five rules that instruct the browser to store including a custom attribute. For each tag, we set the value the reference type in the target, which is mapped to each of each attribute to 'x' and add the JavaScript code that known DOM Clobbering instance. The rules are Named checks whether the markup clobbers the target 'x'. Then, Access on Window ([56] §7.3.3), DOM Tree Accessors we generated markups for object properties 'x.y' and 'x.x' ([56] §3.1.5), Form Element ([56] §4.10.3), Iframe srcdoc combining all pairs of the 142 HTML tags considering three attribute ([56] §4.8.5), and HTMLCollection ([57] §4.2.10.2), relations: sibling tags, parent-child tags, and the srcdoc which we labeled as R1 to R5, respectively. The rest of this attribute value. The experiments with a single tag showed section details each group of clobbering markups and the that only name and id attributes create named properties. rules abused by them. Accordingly, to reduce the number of test cases to a testable Named Access Window. These group of markups leverage size, the generation of markups for object properties did not a single HTML element whose id or name is set to a consider combinations of all HTML attributes, but only those target variable 'x', clobbering window.x due to browsers' of the name and id, e.g., id=x, or id=x, name=y. compliance with the Named Access on the Window Object After generating all markups, we put each of them in a rule (R1) [42]. We reviewed this rule in §2.1. Note that we test webpage, along with a JavaScript code that verifies if use window.x and 'x' interchangeably because all global

4 . Name Rule(s)	Target	Reference	Type	Tag 1	Tag 2	Attribute 1	Attribute 2	Relation	Total			
Reference	Named	Access	Window	R1	win.x, x	WindowProxy	iframe	- name=x - -	1 [27, 42, 55]			
win.x, x	HTMLElement	TS1, TS2	- name=x - -	5 [15, 17, 27, 42]	win.x, x	HTMLElement	any	- id=x - -	141 [14, 21, 22, 27, 42]			
DOM Tree Accessors	R2	doc.x	WindowProxy	iframe	- name=x - -	1 [21, 43, 54]	doc.x	HTMLElement	TS1, TS2	- name=x - -	5 [15, 43]	
doc.x	HTMLElement	object	- id=x - -	1 [43]	doc.x	HTMLElement	img, image	- id=x, name=any - -	2 [15, 17, 43]			
Form	Parent-Child	R3, R1, R2	win.x.y, doc.x.y	HTMLElement	form	TS2, TS3	id=x	k name=x	id=y	k name=y	child	36 [15, 17, 18, 20, 21]
Nested Window	Proxy	R4, R1, R2	win.x.y, doc.x.y	WindowProxy	iframe	iframe	name=x	name=y	srcdoc	attr.	1 [21, 23, 54]	
HTMLCollection	R5, R1, R2	win.x.x	HTMLCollection	any	any	id=x	id=x	child, sibling	141 [14, 22, 57]			
doc.x.x	HTMLCollection	TS2	TS2	id=x	id=x	child, sibling	3 [14, 43, 57]	win.x.y				

HTMLCollection any any id=x, name=y id=x child, sibling 141 [14, 21, 22, 39, 54, 57]
doc.x.y HTMLCollection TS2 TS2 id=x, name=y id=x child, sibling 3 [14, 43, 57] Legend:
R1= Named Access on Window Rule ([56] §7.3.3); R2= DOM Tree Accessors Rule ([56]
§3.1.5); R3= Form Element Rule ([56] §4.10.3); R4= Iframe srcdoc Rule ([56] §4.8.5); R5=
HTMLCollection Rule ([57] §4.2.10.2); win=window; doc=document; TS1=form, embed;
TS2= object, img; image; TS3=button, fieldset, input, output, select, textarea.

TABLE 1: Overview of known DOM Clobbering markups grouped by their corresponding rules in the HTML [56] and DOM [57] specifications.

variables belong to the global window object by default. Post-processing of Results. As the manual review of 31K DOM Tree Accessors. The markups of this group can individual instances is infeasible, we group instances by shadow document properties because browsers comply similar features. We start with preliminary groups based with the DOM Tree Accessors rule (R2) [43], which instructs on the set of browsers they work in and the target they browsers how to retrieve properties of the document object clobber. Then, we look at the structural features, i.e., tag1, (e.g., DOM elements). Similarly to the previous group, these tag2, attribute1, attribute2, and relationship, and we merge markups use a single named HTML element (e.g., object, two groups when all the structural features but one are the or embed) to clobber a property 'x' of the document. same. Accordingly, we reduced the 31K instances to 74 classes, as shown in Table 2, and map each class to our Form Parent-Child Relationship. These markups clobber systematization of known instances. In summary, out of the properties 'X.y' where 'X' can be any of 'x', window.x, 74 classes, 10 classes rely on the Window Named Access, and document.x. First, they exploit either the rules R1 or four classes on DOM Accessors, 13 classes on the Parent- R2 to clobber the base object 'X'. Then, they use the Form Child Relationship, four classes on Nested Window Proxies, Element rule (R3) to clobber property 'y' of object 'X', and finally 43 classes leverage HTMLCollections. i.e., the form elements' parent-child relationships where the browser creates a property of the second element for the Findings. By comparing the 74 DOM Clobbering classes in first element's accessor variable [21]. DOM Clobbering code Table 2 with the 13 previously identified classes in Table 1, that rely on this technique comprise a form tag and a child we discovered that the 31,432 DOM Clobbering markups (e.g., an input) named 'x' and 'y', respectively. include 148 new instances, 481 previously known ones, and 30,803 variants of the known ones, which rely on one of the Nested Window Proxies. These markups use the Iframe five DOM Clobbering techniques of §4.2.1. srcdoc rule (R4) to create nested window proxies that The variants derive from markups that are already known are named with 'x' and 'y', respectively. Similarly to the for DOM Clobbering according to Table 1, but now have previous group of markups, it uses the rule R1 or R2 to one or more additional attributes, or are permuted in part clobber the base object. Then, the stacked iframes enable with a different HTML tag. For example, HTMLCollections attackers to exploit frame navigation features to clobber clobbering window properties may be formed not only for object properties like 'x.y' [21, 23]. two similar HTML tags as in Table 1 (e.g., two a tags with HTMLCollection. The last gour groups of markups rely id=x), but also for certain combinations of dissimilar tags on a different rule known as HTMLCollection (R5). Specif- (e.g., svg and a), which accounts for a large number of ically, when two or more elements have the same id in the clobbering

instances. Other variants are cases where the DOM tree, browsers create an array-like object called additional id and name attributes are added to the existing HTMLCollection [14, 64], which contains all elements clobbering markups. For example, when looking at form with the same id. Elements inside HTMLCollections can be elements and their children in Table 1, we observe that each accessed by (i) their index in the collection and (ii) their tag of the markup has only one id or name. However, as id and name, enabling attackers to abuse R5 to clobber demonstrated by the results in Table 2, these attributes may arrays [21] and loop elements (e.g., 'x' and 'x[i]') as well exist simultaneously on HTML tags and with similar or as object properties like 'x.x' and 'x.y' [22]. Similarly to dissimilar values, resulting in additional clobbering variants. the previous techniques, rules R1-2 can be combined with In comparison, the new clobbering instances rely on new R5 to clobber nested object properties like window.x.y . (pairs of) HTML tags and attributes that were previously not 4.2.2. Clobbering Variables and Object Properties known to be applicable for DOM Clobbering. We observed that 28 out of the 74 identified classes contain at least one Our browser testing experiments uncovered 31,432 distinct new instance, with a total of 148 new instances. From these, DOM Clobbering markups that work in at least one browser, 22 classes contain only new instances (i.e., 142 instances). as summarized in Table 2, from which 145 clobber a variable In the remaining of this section, we briefly describe the new 'x', and the remaining 31,287 clobber 'x.y' and 'x.x'.

5 . Chrome Firefox Opera Edge Safari TB SI UC

95.0.4638

92.0.4515

65.2.3381 82.0.4227

95.0.1020 96.0.1054 95.0.1020 94.1.2

14.7.1 11.0.1

15.0.6 13.3.8 3.2.3 96.0

95.0 39.0

15.1	14.1	13.1	HTML Markup	Clobbered Tag	1	Tag 2	Attribute 1	Attribute 2	Relation	Total
New	z	z	z	z	z	z	z	z	z	z
Named	Access	Window	win.x	TS2	-	id=x	-	-	106	E win.x
customtag,iframe,TS5	-	id=x	-	-	8	1	win.x	TS6,bdi,bdo,big	-	id=x - - 6 win.x
TS4,embed,form	-	n=x	-	-	5	win.x	video,wbr,xmp	-	id=x - - 3 win.x	aside,audio,b - id=x - - 3
E win.x applet	-	n=x	-	-	1	1	win.x	iframe	-	n=x - - 1 win.x
base	-	id=x	-	-	1	win.x	article	-	id=x - - 1	

DOM Tree	Accessors	doc.x	TS4,embed,form	-	n=x	-	5	E doc.x	applet	-	id=x	k	n=x	-	2	2	E
doc.x	iframe	-	id=x	k	n=x	-	-	2	1	doc.x	object	-	id=x	-	-	1	

Form	Parent-Child	win.x.y	form	TS3,TS4	-	fieldset	id=x	k	n=x	(& id=y)	id=y	k	n=y	child	64
doc.x.y	form	TS3,TS4	n=x	(& id=y)	id=y	(& n=x	k	n=y)	child	36	win.x.y	form	TS3,TS4	id=x	
(& n=y)	id=x	& n=y	child	18	win.x.y	form	TS3,TS4,embed	n=x	id=y	& n=x	child	10	doc.x.y		
form	TS3,TS4,embed,	form	n=x	id=y	& n=x	child	10	win.x.x	form	TS3,TS4	id=x	id=x	&		
n=y	child	9	win.x.y	form	button	id=x	k	n=x	(& id=y)	id=y	k	n=y	child	8	win.x.x
n=x	id=y	& n=x	child	6	doc.x.x	form	TS3,TS4	n=x	id=y	& n=x	child	6	E doc.x.x,	win.x.x	

form TS4,embed n=x id=y & n=x child 4 1 E doc.x.y form iframe n=x id=y & n=x child 1 1
E win.x.y, doc.x.y form TS4,embed id=y & n=x id=y & n=x child 4 1 E win.x.y form applet
n=x id=y & n=x child 1 1

Nested Window Proxy doc.x.x iframe iframe n=x id=y & n=x srcdoc 1 E doc.x.y iframe
iframe n=x id=y k n=y srcdoc 2 1 win.x.x iframe iframe n=x id=y & n=x srcdoc 1 E win.x.y
iframe iframe n=x id=y k n=y srcdoc 2 1

HTMLCollection win.x.y TS1,svg,customtag TS1,plaintext id=x id=x & n=y sibling 787
win.x.y TS1,customtag – TS7,iframe TS1,plaintext id=x id=x & n=y child 774 win.x.y
abbr,dl,dt TS13 id=x id=x & n=y child, sibling 274 win.x.y abbr,dl,image,img
TS8,TS12,TS20 id=x id=x & n=y child, sibling 392 win.x.y TS18 TS13,TS14 id=x id=x &
n=y child, sibling 7,480 win.x.y address,dir,dt TS15 id=x id=x & n=y child, sibling 338
doc.x.y applet TS4,applet id=x id=x & n=y child 4 E doc.x.y TS4,applet,embed,form,iframe
TS4,applet,embed,form,iframe n=x id=y & n=x sibling 13 13 E doc.x.y
applet,embed,form,image,img TS4,applet,embed,form,iframe n=x id=y & n=x child 11 11
doc.x.y applet,object TS4,applet id=x id=x & n=y sibling 5 win.x.y dir,div,dt,element TS16
id=x id=x & n=y child, sibling 252 win.x.y div TS17 id=x id=x & n=y child, sibling 66
win.x.y div,dl TS12 id=x id=x & n=y child, sibling 186 win.x.y element,em,embed,fieldset
TS1,plaintext-iframe id=x id=x & n=y child, sibling 876 E win.x.y embed TS4,embed,form
n=x id=y & n=x child, sibling 10 10 E doc.x.y TS4,embed,form,iframe
TS4,embed,form,iframe n=x id=y & n=x sibling 11 11 E doc.x.y TS4,embed,form
TS4,embed,form n=x id=y & n=x sibling 25 25 E doc.x.y embed,image,img iframe n=x
id=y & n=x child 3 3 E doc.x.y embed,image,img T3,TS4,embed,form n=x id=y & n=x
child 15 15 win.x.y TS9,iframe TS1,plaintext-iframe id=x id=x & n=y sibling 1,436 win.x.y
TS9 TS1,plaintext-iframe id=x id=x & n=y child 1,301 E win.x.y form,image
TS4,embed,form n=x id=y & n=x sibling 7 7 E win.x.y TS4,form applet n=x id=y & n=x
sibling 4 4 E win.x.y image embed,form n=x id=y & n=x child 2 2 E win.x.y image,img
TS4,embed,form n=x id=y & n=x child, sibling 16 16 E win.x.y TS4 applet n=x id=y & n=x
child 3 3 win.x.y ins content,data id=x id=x & n=y child, sibling 4 win.x.y TS7, TS8
TS1,plaintext – iframe id=x id=x & n=y sibling 8,848 win.x.y TS8 TS1,TS11,plaintext –
iframe id=x id=x & n=y child 7,526 E doc.x.x, win.x.x object TS4,embed,form n=x id=y &
n=x child 5 5 doc.x.y object TS4 id=x id=x & n=y sibling 3 E doc.x.y object
form,image,img n=x id=y & n=x child 3 3 E doc.x.y object iframe n=x id=y & n=x child 1
1 doc.x.y object image,img id=x id=x & n=y child 2 E doc.x.y object embed,object n=x
id=y & n=x child 2 2 E win.x.y object TS4,embed,form n=x id=y & n=x child, sibling 1 1 E
win.x.y, doc.x.y object TS4,embed,form id=y & n=x id=y & n=x child 5 5 win.x.y svg
iframe id=x & n=y id=x & n=y sibling 1 win.x.y svg TS1,plaintext id=x & n=y id=x & n=y
sibling 125 win.x.y svg,table TS1,plaintext – TS19 id=x & n=y id=x & n=y child 157
win.x.x table iframe id=x id=x & n=y child 1 win.x.x table TS1,plaintext,svg – TS10 id=x
id=x & n=y child 119 0 win.x.y table iframe id=x & n=y id=x & n=y child 1 Total 31,432
148 59 59 46 35 35 46 59 59 44 59 59 43 38 45 52 37 35 59 59 Legend: win=window;
doc=document; TS i= Tag Set in Table 7 of §A.2; n= name; (&p)= optional property p; – =
minus operator; TB= Tor Browser; SI= Samsung Internet; UC= UC Browser; = clobbered; =
clobbering fails;

TABLE 2: Overview of DOM Clobbering markups. Rows marked with E are classes that contain new DOM Clobbering instances. For all rows, clobbering window.x also implies clobbering the variable x. Browsers with similar behaviours are grouped with the same color. The table highlights a total of 10 distinct groups of browser behaviours with respect to DOM Clobbering.

6 instances within each DOM Clobbering technique. amount of DOM Clobbering risk is associated with using Named Access Window and DOM Tree Accessors. We browsers like Firefox Desktop/Android and Chromium-based discovered that any custom HTML tag (e.g., customtag) browsers on Desktop/Android in which 35 and 59 classes can be used to clobber a target variable x and window.x in all of DOM Clobbering markups work, respectively. web browsers. Also, iframe tags with id=x can clobber 4.2.3. Clobbering Native APIs document.x and named applet elements can clobber both window.x and document.x. In total, we found five new Overall, we identified a total of 347 DOM APIs (Table 12) instances across four out of the 14 classes that rely on that can be clobbered in at least one browser using one of the Window Named Access and DOM Accessors techniques. the markups of §4.2.2, including 233 document and 114 window APIs. We observed that all document methods Form Parent-Child. We discovered that browsers like and properties except the location property (i.e., 233 Firefox and Safari create accessor properties on JavaScript APIs) can be clobbered in all browsers unanimously, as objects due to element’s ancestral relationship in the DOM expected by the named property visibility algorithm [45] of tree for previously unknown pairs of tags and attributes, such the specification [19, 56]. However, this experiment resulted as a parent form tag with a embed, iframe, or applet in a new finding that for a total of 114/347 window APIs child with both a name and id attribute. Overall, among the (i.e., 91 properties and 23 methods), named properties can 13 classes that rely on elements’ parent-child relationships, shadow native properties that would otherwise appear on the we found four new markups in four different classes. object in at least one browser, resulting in DOM Clobbering. Nested Window Proxy. We identified two new clob- This includes security-sensitive APIs such as the cache bering markups in two out of the four classes which use storage [34], notification API [67], trusted types [35], and the Nested Window Proxies technique. In particular, we web storage [68]-to name only a few instances. The complete discovered that using the id attribute in the nested frames list of clobbered window methods and properties is in creates a named property on the base frame, referring to Table 13 of §A.2. We observed that for 57/114 clobbered a WindowProxy, whereas id on the base frame does APIs, there is at least one browser that disagrees with others. not create a WindowProxy accessible through the global window or document. 5. Detection and Prevalence HTMLCollection. We found 137 new clobbering in- The second part of this paper intends to evaluate the impact, stances (across 18 classes) that lead to the construction prevalence and variety of DOM Clobbering vulnerabilities of HTMLCollections in a different way. Specifically, we in real-world web applications (RQ2 of §3). In §5.1, we first discovered that some browsers (e.g., Chrome and Firefox) present TheThing, an automated DOM Clobbering detection create an HTMLCollection not only when two elements tool. Then, in §5.2, we present our experiment results. share the same id, but also when they have the same name value. However, we observed that this happens only for 5.1. Detection certain (combinations of) HTML tags,

e.g., two object We formulate the problem of detecting DOM Clobbering tags and two form tags with the same name can form an vulnerabilities into a series of data flow analysis tasks HTMLCollection, but not two div tags. where we identify clobberable JavaScript variables, object Analysis of Browsers' Behaviours. Our experiments re- properties, and native APIs whose value ultimately reach vealed that browsers exhibit divergent behaviours when link- security-sensitive instructions, such as script src and eval. ing named HTML elements to JavaScript variables (Table 2). Identifying such data flows via pure static analysis is For example, we observed that for a significant fraction of not an easy task given the dynamic nature of client-side the clobbering markups (i.e., 31,243 out of 31,432), there JavaScript programs [32, 69, 70] and the scale of the analysis is at least one browser that disagrees with others, rendering as studying DOM Clobbering vulnerabilities requires the the task of defending against DOM Clobbering increasingly collection and analysis of hundreds of webpages of real web more challenging. In summary, we identified 10 distinct applications. Accordingly, we use and extend state-of-the-art groups of browser behaviours with respect to different DOM property graphs for JavaScript and graph traversals [32] to Clobbering markups, which are highlighted in Table 2 in col- identify potentially-vulnerable data flows and then use forced ors, showing that while most of the attacks are shared across execution to confirm the presence of the vulnerability. browsers, many others only work with specific browsers. Figure 1 shows the architecture of TheThing. At a high The table shows that all Safari and iOS-based browsers level, it has three main components: (i) a web crawler have their own distinct behaviours, whereas browsers like to collect webpages' data and the JavaScript code, (ii) a Chrome, Opera, and Edge on Desktop and Android exhibit vulnerability analysis component that uses property graphs the same behaviour. Note that, in general, similarities in and traversals for identifying potential DOM Clobbering behaviours are expected because some browsers rely on sources and capturing data flows to security-sensitive sinks, the same underlying engine. For example, Chrome, Edge and finally (iii) a vulnerability verification component that and Opera on Desktop are all Blink-based browsers [65], dynamically confirms the candidate data flows by instru- whereas iOS browsers are required to use the WebKit engine mentioning the code and forcefully executing it in a browser of Apple [66]. Finally, we observed that the least and highest to check if the flow can occur at runtime. The rest of this section details each component.

7 Figure 1: Architecture of TheThing.

5.1.1. Data Collection hoisting [41] and double-clobbering [17].

For example, a runtime error in one script causes the browser to stop parsing To collect the client-side code of web applications, we that script, and continue with parsing of the rest of the developed a JavaScript-enabled crawler leveraging Pup- scripts. Hence, variables initialized in the first script are peteer [71] and Chrome DevTools Protocol (CDP) [72]. treated as undefined and can be a candidate DOM Clobbering Starting from a seed URL of the website under test, it visits source. Such runtime errors can be caused intentionally by the webpages following a depth-first strategy, and stops when attackers by a preliminary clobbering, e.g., clobbering a it doesn't find new URLs, or the maximum of 100 URLs native DOM function that is invoked in a script shadows its is reached. During the visit, it collects the page resources value to an HTML element, which is not callable, leading (e.g., scripts) and runtime state

values (i.e., fired events and to a runtime error (Cf. Table 3). Accordingly, we changed DOM objects' properties) using the CDP and Puppeteer. the normalization procedure to keep track of the script of 5.1.2. Vulnerability Analysis origin for each AST node. Second, the semantic types of JAW are tailored for client- Given the webpages' data collected by the crawler, side CSRF vulnerabilities and are not sufficient to model TheThing creates a property graph of the client-side DOM clobbering. Accordingly, we added a new set of generic JavaScript program leveraging a modified engine of JAW [32]. semantic types for DOM Clobbering sources (Cf. Tables 3 Then, we formulate the problem of finding potential DOM and 9) and security-sensitive JavaScript sinks (Cf. Table 8). Clobbering data flows into a series of graph traversal queries. Third, JAW does not fully support ES6, resulting in impre- Hybrid Property Graphs. HPGs are graph-based repre- cise control and data flow models. Accordingly, we applied sentations of client-side JavaScript programs that unify several enhancements. For example, we added support to multiple static code representations and runtime state values. bind the function call arguments to their definition parameters State values are event traces and environment properties, when the code uses the ES6 Rest parameters [76] and the e.g., the values of cookies and web storage. The static Spread operator [77] which improves the precision of the code representation comprises several graphs, e.g., Abstract call graph and PDG edges. Also, we created bindings for Syntax Tree (AST), Control Flow Graph (CFG) and Program the this object depending on the calling context [78], Dependence Graph (PDG) that model the nesting of the and the binding for the arguments object for non-arrow syntactical constructs of a program, the order and conditions functions [79] to improve pointer analysis tasks. for the execution of program instructions, and the data flow Analysis Traversals. After construction of an HPG, we and control dependencies within the statements of a program, traverse it to identify DOM Clobbering source nodes in the respectively. HPGs also model the event-driven transfer of graph. Table 3 presents the various types of DOM Clobbering control within JavaScript programs via the Event Registration, sources and their properties. The table shows that clobberable Dispatch and Dependency Graph [32]. Finally, they include native DOM APIs discovered in §4.2.3 can act as a DOM Semantic Types, which are labels initially assigned to source Clobbering source. Identifying these objects in the program and sink nodes to capture the semantic of those instructions is a matter of searching for a pre-defined syntactic structure, and then propagated through the graph following the program which is similar to other taint-style vulnerabilities like client- calculation. These representations are encoded in a directed side XSS. However, contrary to the traditional taint analysis, graph in which nodes and edges can have labels and key- not all DOM Clobbering sources are pre-defined syntactic value properties, known as a labeled property graph [32, 73]. objects. Instead, they can be a specific property of a program, Model Construction. After collecting the webpages' scripts identifying which requires tracking the propagation of data and state values, TheThing instantiates an HPG, and imports flows within the program itself. This is because any used it into a Neo4j docker instance [74], allowing the graph to be variable that is undefined within its execution context (i.e., traversed declaratively using the Cypher query language [75]. previously not declared and assigned) can act as a DOM Unfortunately, we could not use JAW as-is and modified it Clobbering source. To identify such sources, we use PDG to

address several of its shortcomings. First, when building a data dependency edges, which specify that a variable defined graph, JAW normalizes the webpage code by combining code at a source node is subsequently used at the destination inside script tags into a single script. However, identifying node. Specifically, we query the graph for Identifier nodes DOM Clobbering sources may require to distinguish the containing a variable v with no incoming PDG edge from code across two different scripts due to JavaScript variable any AssignmentExpression or VariableDeclaration nodes that

8 . Object DOM Clobbering Source When? Threat # Sinks # Flows # Conf. # Pages # Sites
 $v \in NP$, CLOB(v) Client-side XSS 37,941,540 3,688 3,677 1,572 474 S2: $v \in / NP$, v and window. v are not assigned before, v is not Request Forgery 2,555,147 1,406 1,403 541 398 declared with var, let and const before Storage Manipulation 1,047,512 1,369 1,365 418 382 Open Redirect 1,306,603 1,228 1,227 391 385 window. v S3: $v \in NP$, CLOB(v) JSON Injection 9,610,162 793 793 345 343 S4: $v \in / NP$, v and window. v are not assigned before, v is Cookie Manipulation 1,702,340 266 266 204 195 not declared with var afterwards within the same script, or Websocket Hijacking 21,252 367 367 183 147 anywhere before RegEx Injection 13,325,791 284 284 98 98 S5: $v \in / NP$, v or window. v is assigned or declared with any of Doc. Domain Manip. 55,266 85 85 69 69 the var, let and const keywords within any previous script postMessage Manip. 119,971 0 0 0 0 that contains an invocation of function f such that $f \in NP$, File Read Path Manip. 57,789 0 0 0 0 CLOB(f) Total 67,743,373 9,486 9,467 3,821 491 document. v S6: $v \in / NP$ S7: $v \in NP$, CLOB(v) Legend: Conf.= Dynamically Confirmed Legend: NP= native property; CLOB(v)= v is a clobberable NP based on §4.2.3. TABLE 4: Prevalence and impact of DOM Clobbering in Tranco top 5K TABLE 3: Description of properties of DOM Clobbering sources. sites. The table shows the number of clobberable data flows to security sensitive sinks of Table 8, the number of affected webpages, and websites. assign to or declare the variable v . If there is such PDG the source location. edge, we further check whether the declaration/ assignment As the next step, we confirm the existence of the data statement can hinder the clobberability of v based on the flow to the sink instructions. To do that, we first taint each criteria in Table 3, which can depend on the declaration clobberable source, execute the program by loading it via scope (i.e., same script or not), declaration position (i.e., Puppeteer, and check if we can observe the data flow reported before or after), and the declaration keyword (e.g., var vs by the static analyzer. If that is not the case, we forcefully let) of that statement. execute the path toward sinks to check if there is an execution After identifying the source nodes, we associate to each of the program in which the data flow to the target sink occurs. of them a label that captures the semantic type of the We use forced execution to find candidate pages among those source, e.g., a clobberable native property or custom variable where Puppeteer could not connect sources with sinks, and (Cf. Table 9). Then, given a list of JavaScript sinks, we later validate the presence of the vulnerability manually. identify each of them in the graph and assign each a Specifically, for each branch in the path control flow, we relevant semantic type. Semantic types assigned to sink forcefully execute the program once for the true and once for instructions are propagated to other functions that encapsulate the false branch, until we hit a execution path with the target the same semantic, e.g., the type WIN_LOC_WRITE is set data flow, or we exhaustively checked possible execution for instructions that set the value

of `window.location`, `paths`. We observed that the number of branches between such as `window.location.replace()`, and is then DOM Clobbering sources and sinks is in practice small (i.e., propagated to all other developer-defined functions that can less than 10), as we will show in §5.2. Finally, as forced set its value through one of their parameters. TheThing execution may also lead to spurious execution paths, we considers different sink types to enable us to capture the manually validate the decision reported by TheThing and potential consequences of DOM Clobbering. The complete examine the exploitability. list of sinks is in Table 8, which is derived by surveying and aggregating the JavaScript sinks considered in prior academic 5.2. Prevalence in the Wild and non-academic resources (see, i.e., [1, 5, 36, 37, 80– 90]). Finally, we conduct forward data flow analysis by We quantified the prevalence and impact of DOM Clob- propagating semantic types from sources to sinks, and select bering on the top 5K websites using the Tranco list [91] those flows where a node with a sink semantic type is of Nov 1st, 2021 (ID: Y3JG), where we first selected the tainted with a source type (i.e., pick up the attacker-controlled top 5K domains by excluding the duplicates like local values). The concrete queries are presented in Table 11 of versions of websites (e.g., `google.com` vs `google.de`), and §A.2. This component outputs a set of paths with potential then instantiated TheThing for each of the them. data flows from a DOM Clobbering source to a sink. Data Collection Statistics. Starting from the 5K seed URLs, TheThing collected 205,696 webpages, ranging between 1 to 5.1.3. Vulnerability Verification 91 pages per site (41 pages on average). Out of the 205,696 Given a set of potential DOM Clobbering data flows, webpages, 187,280 are unique pages based on their set of the goal of this step is to verify each flow and eliminate scripts. From the 187K pages, TheThing extracted 18,351,815 potential false positives. To accomplish this goal, TheThing scripts with a total of 24,664,686,928 LoC. Accordingly, features a light-weight, in-browser dynamic taint analysis TheThing generated 187,280 HPGs by processing an average engine leveraging Iroh.js [33]. After instrumenting the code of 98 scripts and 131,700 LoC per page. with Iroh for dynamic analysis, we first check whether the Vulnerability Prevalence. The analysis of 187,280 HPGs source variable of the data flow is clobberable by creating a resulted in the identification of 20,580,350 DOM Clobbering suitable HTML clobbering payload for that variable using sources and 67,743,373 sinks, which amounts to an average the DOM Clobbering classes of §4. We inject the payload of 110 sources and 362 sinks per webpage. Out of these, to the DOM tree and subsequently verify the clobberability static analysis revealed a total of 9,486 potential data flows of the source variable by dynamically logging its value at from the sources to the sinks, from which the majority (i.e.,

9 9,467) were confirmed dynamically. We observed that these the high number of affected webpages, we randomly se- vulnerable data flows affect around 2% of the webpages (i.e., lected two vulnerable pages per each of the 491 affected 3,821 out of 187,280) and 9.8% of the tested websites (i.e., sites, and subsequently checked whether we can insert a 491 out of 5K) in total. Table 4 summarizes our findings. DOM Clobbering markup in the page by leveraging the Vulnerability Impact. We observed that the 9,467 vulner- functionalities offered by the application, or through URL abilities can have different security implications, as shown parameters, which could allow us to overwrite the clobberable in Table 4. The most common consequence is XSS that variable identified by TheThing. To be able to use

protected accounts for around 38.8% of the vulnerabilities, whereas functionalities offered by the websites (e.g., creating posts, the least common consequence is document domain manipulation: adding comments, etc) and also prevent any side effects for manipulation [83, 84] that corresponds to less than 1% of other users, we created our own test accounts for 358 sites the total vulnerabilities. Other common consequences were that supported this feature without monetary costs, and for client-side state manipulation (17.2%), client-side request the rest, we limited our tests to the public functionalities (e.g., forgery (14.8%) and DOM-based open redirection (12.9%). search) without persisting any data. As a result, we created Finally, the remaining 15.3% of vulnerabilities had other a proof-of-concept exploit for 44 websites in total, affecting repercussions like JSON injection and Websocket connection popular sites and functionalities like Trello boards, Wiki hijack. We provide more information on each of these threats pages in WikiBooks and WikiDot, comments in Vimeo and in Table 8 of §A.2. VK, reviews in TripAdvisor and OpenTable, posts in Fandom and JustPaste, surveys in SuveryMonkey, poster designs in Verification and False Positives. Considering the high PosterMyWall, and finally item searches in GitHub Shop, number of reported data flows by the static analyzer (Cf. AliExpress, AliBaba and Telam News-to name only a few §5.1.2), it was infeasible to verify all of them manually. examples. The exploits enable an attacker to achieve XSS, Instead, we followed a semi-automatic approach leveraging open redirect, and client-side request forgery in 35, five, and a combination of dynamic analysis, forceful execution and four sites, respectively. We refer interested readers to §A.1 manual analysis, as detailed in §5.1.3. for a few case studies of the confirmed attacks. We observed that in a large number of cases (46.1%, i.e., 4,373 flows), the dynamic verification component can 6. Defenses successfully confirm the existence of the vulnerability by loading the page and executing it via Puppeteer, whereas This section addresses RQ3 of §3. First, in §6.1, we have in the remaining cases (i.e., 5,113 flows), it needs to force a critical look at the existing countermeasures and evaluate execute between one to ten conditional branches (four on their robustness and cost-benefit tradeoff leveraging what we average) before it can confirm or reject the data flow and learned from Sections 4 and 5. Then, in §6.2, we analyze the terminate. As a result of this process, the verifier eliminated common mistakes of the 491 vulnerable sites (see §5), and a total of 19 FPs across 11 of the 491 vulnerable sites, and distill a list of recommendations and secure coding patterns confirmed the rest (i.e., 5,094 flows within 2,643 webpages of that can resolve those issues. 491 sites). We manually verified and investigated the reason for each FP, and discovered that eight FPs occur during the 6.1. Evaluation of Existing Countermeasures data flow analysis for identification of DOM Clobbering Disabling DOM Clobbering Features. DOM Clobbering sources, and 11 during the data flow analysis from sources can be solved by disabling named properties [19, 25, 27]. to sinks. The former cases happened because a variable According to Chrome telemetry [28], disabling named prop- was declared or assigned using a dynamic code generation erties for clobbered variable accesses could break ~10.5% construct for which the statement nodes and PDG edges of the webpages. Our results of §5.2 are in line with these were missing in the HPG, and the latter cases occurred due numbers, and we observed that 13.3% of the webpages use to dynamically fetched code where the value of the tainted at least an instance of clobbered variable accesses. variables changed,

inaccurate pointer analysis for dynamic As webpages tend to reuse code via shared scripts, a patch property lookups, and removal of event handlers that changed in a script may fix multiple websites. Accordingly, using the tainted variables. number of webpages may not accurately quantify the cost Finally, we manually validated the feasibility of the of fixing breakage. As an alternative, we can measure the forcefully executed data flows by randomly selecting two number of affected websites, and our results show that the pages per site, from the 2,643 pages of the 491 websites affected pages do not concentrate on a small number of sites, whose data flows were confirmed by forced execution. Our but they scatter over 51.2% of the top 5K sites. random sampling included 491 sites, 982 pages and 2076 While breakage adequately measures the cost of this data flows, out of which we could not determine a realistic solution, it may not be a good indicator for the actual execution path for at least 42 data flows in 42 sites, leaving benefits, i.e., fixed websites. Our results show that 118 us with 2,034 vulnerable data flows of 491 websites. websites of 2,561 potentially broken sites will be fixed, which is about 4.61% of the broken websites (and 2.4% of 5.3. Confirming Exploitability of Vulnerabilities the total). However, our results also show that a large fraction We manually examined whether the identified vulnera- of vulnerable websites are not considered by breakage. In bilities can be effectively exploited by an attacker. Given particular, we found 373 websites (76% of the vulnerable

10 Default DOM Clobbering markups identified in §4 to each of them,

Strict HTML Sanitizer Ò Bypassed Pct. Ref. and for each input vetted whether the sanitizer removes Client-side JS

1. DOMPurify 8.7K 534 49.7K 7.9M 29,995 95.4% [7]

or changes the named properties in the output markup.

1. Google Closure Lib. 4.3K 1K - 117K - - [92] For each sanitizer, we tested both the default and most

2. JS-XSS 4.4K 584 136K 8.7M 25,592 81.4% [93]

3. Sanitize-HTML 2.8K 316 102K 4.7M 79 0.25% [94]

strict configuration that it offers. We marked a sanitizer as

1. Google Caja 1.1K 123 - - 27,951 88.9% [95] vulnerable if there is at least one clobbering markup that

Node.js bypasses the sanitizer without being altered. Finally, we

1. Insane 394 21 - 55.3K 5 0.02% [96] marked sanitizers as partially vulnerable when they encode

2. Bleach 117 19 - 1.6K 2,288 7.2% [97]

3. Angular-sanitize 100 237 49.1K 936K - - [98] the < and > symbols of HTML tags but do not remove

4. Yahoo html-purify 40 6 - 708 28,807 91.6% [99] or change the DOM Clobbering named properties because

5. Arcgis 11 2 - 32.6K - - [100]

encoding these symbols would not help when applications Python

1. Mozilla Bleach 2.3K 230 155K 17.5M 31,132 99.05% [101]

expect inputs in an HTML format.

1. LXML 2K 481 216K 29.9M 28,211 89.7% [102] Table 5 summarizes our findings. In total, we observed
2. HTML Sanitizer 61 19 - 17.9K 332 1.06% [103] that 16 and 13 out of 29 sanitizers are vulnerable to at
3. HtmlLaundry 27 4 - 1.1K 1,460 4.6% [104]
4. Django-html-sanitizer 20 62 - 2.8K - - [105] least one DOM Clobbering markup in their default and most

PHP strict sanitization configuration, respectively. In both of the

1. HtmlPurifier 2.4K 284 82.7K 2.5M - - [106] configurations, four sanitizers are only partially vulnerable, as
2. Html-sanitizer 333 36 - 30.8K - - [107]
3. Symfony Sanitizer 104 1 - 7 - - [108]

they escape the markup rather than cleansing the named prop-

1. HTMLawed 30 14 - 390K 21,211 67.4% [109] erties. Finally, when looking at the remaining 13 sanitizers,
2. Typo3 Sanitizer 13 10 - 88.9K 23,942 76.1% [110]

we observe that they implement a robust, enabled-by-default C# defense. However, in all cases, they remove named properties

1. AntiXssEncoder 2.6K 1K - 6.4K 31,390 99.8% [111]
2. HtmlSanitizer 1.1K 162 1.8K 108K 654 2.08% [112] unconditionally, i.e., for all input markups including those
3. AJAX Toolkit 275 133 4.2K 264 - - [113] combinations that do not lead to DOM Clobbering, e.g., an
4. NSoup 147 46 - 72 - - [114]
5. HtmlRuleSanitizer 50 16 30 308 - - [115]

anchor tag with name=x does not clobber the variable x. Java While such a strict approach is effective, it may hinder the

1. Jsoup 9.2K 2K 98.4K - - - [116] usability of these libraries in cases where developers need
2. OWASP HTML Sanitizer 647 171 - - - [117] to use id and name attributes for legitimate functionalities.
3. Antisamy 105 72 - - - [118]
4. HtmlCleaner - - - 824 28,951 92.1% [119] Content-Security Policy (CSP). When attackers can clobber

Total Vuln. (+) 16 13 the src attribute of dynamically created scripts, they can Legend: = GitHub Stars; Ò= GitHub Forks; = GitHub UsedBy; = Monthly Downloads; load and execute

arbitrary JavaScript code. In these cases, the = Vulnerable; = Partially Vulnerable; = Not Vulnerable CSP script-src directive [29] can be used to constrain TABLE 5: Robustness of top five HTML sanitizers of web programming the value of script sources to a set of trusted domains, languages against the 31.4K DOM Clobbering instances of §4.2. The preventing attacker-loaded code to be executed [12, 22, 30]. table shows the results for both the default and the most strict sanitizer configurations. The tested sanitizer versions are in Table 10. However, unlike malicious JavaScript injected by the attacker, injected HTML code is not blocked by CSP. Accordingly, ones and 7.5% of the total) that will benefit from such a CSP does not mitigate other variants of DOM Clobbering that solution. Overall, when comparing the cost and benefits, the do not require script src manipulation, e.g., clobbering the ratio of vulnerable over potentially-broken websites is about parameters of dynamic code evaluation constructs like new 1:5.2 (i.e., 491 vulnerable and 2,561 potentially-broken sites). Function() can lead to CSP-bypassable XSS. Our eval- HTML Sanitization. HTML sanitizers can sanitize the input uation in §5.2 shows that 37.7% of the DOM Clobbering markups before adding them to the DOM tree, e.g., by vulnerabilities that lead to XSS (i.e., 1,385 out of 3,677), removing the id and name attributes from certain (combina- which accounts for 14.7% of the total vulnerabilities can be tions of) HTML tags (Cf. §4). To assess the robustness of mitigated by CSP, whereas the remaining ones cannot. the popular HTML sanitizers against DOM Clobbering, we Freezing Object Properties. Another way to mitigate dynamically tested them against all of the DOM Clobbering DOM Clobbering is to freeze DOM objects [51], e.g., via instances we identified in §4. First, we selected the top Object.freeze() method [121], which prevents the five web programming languages based on the GitHub 2021 object to be overwritten by named DOM elements. While Octoverse report [120], i.e., JavaScript, Python, Java, C# effective, determining all objects and object properties that and PHP. We considered both client-side and server-side need to be frozen is a non-trivial, error-prone task for web JavaScript (i.e., node.js). Then, we searched for sanitizers of developers. Also, sealed objects cannot be changed anymore, each language and selected the top five based on their GitHub hindering the dynamic composition of webpages. Finally, stars, forks and UsedBy, and the number of downloads in native properties cannot be frozen, rendering this approach their respective package managers (e.g., npm for node.js, ineffective when the DOM Clobbering source is a clobberable packagist for PHP, etc). This process led to the identification native property, which accounts for ~21.5% of vulnerabilities of 29 HTML sanitizer libraries, as for Java, we identified (i.e., 2,037 out of 9,467) in §5.2. only four sanitizers. After identifying the popular sanitizers, we input the 31.4K

11 .

Code Pattern Description # Flows # Pages # Sites

A VAR1 is not declared or assigned yet, thus window.VAR1 is clobberable. 3,134 1,214 143
 B BA is a clobberable built-in API (§4.2.3), thus BA, window.BA and document.BA are clobberable. 2,037 832 99

C Assignment to document properties is always shadowed by DOM Clobbering (§4.2.3).
1,896 655 81

D VAR1 is declared with let that does not create property on window, thus window.VAR1 is
367 153 18 clobberable.

E VAR1 is initialized without var in the same script and after the sink, but this does not
result in hoisting. 1,635 792 116

F VAR1 is initialized with var, but in a different script and after the sink statement. 121 50
12

G VAR1 is initialized in a script where a built-in method can be clobbered and cause an
error in parsing 53 36 7 that script, hence window.VAR1 can be clobbered in a subsequent
script (double clobbering).

H VAR1 is initialized in a different script as a property of the window or without any
modifiers after the 224 89 15 sink statement, thus window.VAR1 is clobberable. Legend:
BA= Built-in API; WinDoc = Window or Document Object; [code]= Alternative code
statement; Red= Clobberable; Yellow = script 1; Orange = script 2;

TABLE 6: Overview of DOM Clobbering code patterns in the wild. Different background
colors represent code in two different script tags.

6.2. Secure Code Patterns could patch the patterns A, D, E, F, and H. When the value
needs to be used in multiple scripts, as in patterns F and H, Our evaluation of existing DOM
Clobbering counter- the declaration should be in the same (or a previous) script, measures
in §6.1 revealed that they are not sufficient for but not in subsequent ones. complete
protection in a large number of cases. In this section, Strict Type Checking. Another
common mistake en- we have a closer look at the variety of DOM Clobbering abling DOM
Clobbering is treating DOM properties, like vulnerabilities in real web applications (§5.2),
identifying document and window properties as safe, trusted values vulnerable behaviours
and the common types of coding (e.g., patterns B, C, and G). Instead, developers should
extend mistakes. Then, we use these vulnerable behaviours to distill the trust boundary to
these properties, verifying their type a list of recommendations and defensive coding
patterns before using them in security-sensitive instructions, e.g., that developers could
apply to prevent DOM Clobbering. using the instanceof [122] and typeof [123] operators.
To achieve this objective, we extracted the vulnerable lines Do Not Use Document for
Global Variables. Properties of of code and characterized them based on their high-level
document can always be overwritten by DOM Clobbering, syntax and semantics,
identifying eight distinct vulnerable even immediately after they are assigned a value, as
in code patterns in the wild. pattern C. Accordingly, developers should refrain from using
Table 6 summarizes our findings. We observe that the most document as a mean to store
and retrieve global values. common mistakes are patterns A and E, in which the devel-
Instead, they can declare variables with const or var in oper references an undefined
variable through the window the global context, or use the globalThis object [124]. object,
and then use the result in a sensitive instruction, whereas the least common, but also
more complex mistakes Namespace Isolation. While robust sanitizers in §6.1 re- are
patterns F, G and H where the vulnerability originates move named properties, an

alternative solution is to separate due to the position of the instructions that span across two the namespace of variables defined by JavaScript code and different script tags. Other common mistakes are patterns B named properties in user-generated markups. For example, and C, where developers treat custom and native document we observed that the markdown to HTML converter of ap- and window properties as trusted values that can be safely plications like GitHub and BitBucket prefixes id and name used in sensitive operations. The rest of this section presents attribute values of user-generated markup with a specific secure coding patterns that can prevent DOM Clobbering. string. Motivated by this solution, one can monitor runtime changes in the DOM tree via the MutationObserver Explicit Variable Declarations. As shown in Table 6, a key API [125], and prefix named properties of all dynamically element enabling DOM Clobbering is use of the || operator inserted markups before adding them to the tree, which to rely on specific defaults when the primary, intended patches all patterns in Table 6. variable or property is undefined. As an alternative solution, developers can initialize those variables with the default 7. Summary and Discussion value when they are undefined using var declarations, which prevents named properties to overshadow them according to Clobbering Markups Come In Many Forms. In this the named property visibility algorithm [45]. This solution paper, we proposed a systematic technique to identify DOM

12 Clobbering markups, and showed that they come in many fragments of JavaScript code, known as script gadgets, are forms, with a total of 31,432 attack markups that rely on five unexpectedly executed as a result of a non-script markup different techniques, including 148 new instances and 30,803 injected by attackers. The authors used a modified browser new variants. We observed that browsers exhibit divergent engine [1] to measure the prevalence of these gadgets, behaviours when handling named properties. For example, and demonstrated that they are prevalent and can bypass for a significant fraction of the markups (i.e., 99%), there existing XSS mitigations, such as HTML sanitizers [7] and is at least one browser that disagrees with others, making it CSP [29, 30]. Later, Roth et. al. [130] quantified the impact increasingly more challenging to enforce robust defenses. of script gadgets on CSP in the wild. Similarly, Heiderich DOM Clobbering is Ubiquitous. DOM Clobbering vulner- et. al. [6] discovered mutation-based XSS attacks (mXSS), abilities are prevalent, affecting over 9.8% of the top 5K showing how specific browser-based mutations of DOM sites, with the consequences ranging from XSS to user state content and insecure JavaScript that reads and rewrites HTML manipulation, request forgery and client-side open redirects elements can transform initially secure DOM markup to code. in the majority of the cases, i.e., 83.7% (see §5.2). While all these three attacks can transform non-script markup to executable code, the elements enabling DOM Clobbering Defenses Helpful but May not Completely Cut it. The is largely different, i.e., script gadgets rely on event handlers evaluation of existing DOM Clobbering countermeasures (§6) and mXSS attacks abuse innerHTML mutations, whereas suggests that each can only mitigate a fraction of the attacks. DOM Clobbering is the result of a complex interplay of the For example, 55% of the popular HTML sanitizers are default browser behaviors and insecure use of named property vulnerable to at least one of the 31K clobbering markups by accesses in JavaScript programs. Contrary to these works, our default, and CSP cannot mitigate over 85% of the identified study focuses on DOM Clobbering, systematically

testing vulnerabilities. Protecting such a fraction of the attack mobile and desktop browsers, identifying insecure coding surface without switching named properties off completely patterns using both static and dynamic analysis techniques, is a more costly task, requiring developers to be aware of and demonstrating their exploitability. corner case behaviors of browsers and revisit the design and Multiple instances of DOM Clobbering vulnerabilities implementation of their systems, e.g., strict type checking, have been discovered in the last 12 years by both aca- explicit variable declarations, or namespace isolation. demics [7, 13, 131] and security analysts [14, 21–23, 54], Open Science and Website. To support the future research with the first public instance identified in 2010 by Rydstedt effort, we publicly release TheThing [126], the automated et. al. as a way to circumvent frame busters [13]. The term browser testing pipeline [127] that identifies clobbering ‘DOM Clobbering’ itself emerged in 2013, when Gareth markups (see §4), and an interactive version of markups1 . Heyes [14] demonstrated how this class of vulnerabilities can Ethical Discussion. Our experiments on live sites do not escalate to client-side code execution. Due to such nefarious target any real user. Tests requiring to persist data, e.g., consequences of DOM Clobbering, prior academic studies store a markup, is exclusively restricted to user accounts has primarily focused on its defenses (e.g., [7, 26, 51]). that we created on those sites. Also, we excluded testing Most notably, Heiderich et. al. proposed the JSAgents functionalities where we could not control the impact and library [51] and later the DOMPurify sanitizer [7] to mitigate visibility of the injected markup (e.g., publicly accessible the security implications induced by markup injection, such posts and comments). Tests on public functionalities was as DOM Clobbering and client-side XSS [1, 5]. Our research performed without persistently injecting any markup. completes the missing pieces of these works by systematically The vulnerabilities and security risks identified in this studying DOM Clobbering attack techniques, their preva- paper affects 491 websites and 16 sanitizer libraries. We lence, and effectiveness of the existing countermeasures. started the process of notifying the affected parties in March 2022 following the best disclosure practices [128, 129], 9. Conclusion where we prioritized our reports by severity. We sent an initial notification that includes the vulnerability details, or a In this paper, we performed, to the best of our knowledge, proof-of-concept exploit, followed by an additional reminder the first comprehensive study of DOM Clobbering, system- every three weeks to maximize the remediation rate. At the atically covering clobbering techniques, browser behaviours, time of preparing the camera-ready, we have notified all vulnerability prevalence, and defenses. Starting with a com- affected parties at least once, out of which 72 sites have prehensive survey of existing literature and dynamic analysis already confirmed the issues, and 21 sites patched them, such of 19 web browsers, we presented the first taxonomy of DOM as GitHub, Vimeo, Fandom, TripAdvisor and SuveryMonkey. Clobbering, uncovering 31K distinct markups that use five different techniques to clobber JavaScript variables. Then,

1. Related Work we presented TheThing, the first DOM Clobbering detection

Reusing the webpages’ legitimate JavaScript code to obtain tool, and instantiated it on the top of the Tranco top 5K sites, arbitrary client-side code execution have been the subject showing that DOM Clobbering vulnerabilities are prevalent. of several research endeavors in the past. Most notably, Finally, we demonstrated that existing countermeasures are

Lekies et. al. [12] described a new attack where small not sufficient to mitigate a significant fraction of the vulnerabilities, and accordingly proposed several recommendations

1. <https://soheilkhodayari.github.io/DOMClobbering> and secure coding patterns for developers.

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15 Appendix A. Name HTML Tags TS1 a, abbr, acronym, address, applet, area, article, aside, audio, b, base, basefont, bdi, A.1. Case Studies bdo, bg sound, big, blink, blockquote, br, button, canvas, center, cite, code, com- mand, content, data, datalist, dd, del, details, dfn, dialog, dir, div, dl, dt, element, This section reports on a few manually vetted case studies em, embed, fieldset, figcaption, figure, font, footer, form, h1, header, hgroup, hr, i, iframe, image, img, input, ins, isindex, kbd, keygen, label, legend, li, link, listing, of the confirmed attacks. We note that the affected parties main, map, mark, marquee, menu, menuitem, meta, meter, multicol, nav, nextid, nobr, have been promptly informed of the vulnerability, and have noembed, noframes, noscript, object, ol, optgroup, option, output, p, param, picture, pre, progress, q, rb, rp, rt, rtc, ruby, s, samp, script, section, select, shadow, slot, small, already patched them (see §7). source, spacer, span, strike, strong, style, sub, summary, sup, table, template, textarea, time, title, track, tt, u, ul, var, video, wbr, xmp GitHub. This vulnerability affects the GitHub Shop and orig- TS2 blockquote, br, button, canvas, center, cite, code, command, content, data, datalist, inated when loading the Boomerag JavaScript library [132]. dd, del, details, dfn, dialog, dir, div, dl, dt, element, em, embed, fieldset, figcaption, In more details, the code followed the vulnerable pattern G figure, font, footer, form, h1, header, hgroup, hr, i, image, img, input, ins, isindex, kbd, keygen, label, legend, li, link, listing, main, map, mark, marquee, menu, menuitem, of Table 6, where a variable called BOOMR was defined in meta, meter, multicol, nav, nextid, nobr, noembed, noframes, noscript, object, ol, an inital script that contained a clobberable, invoked native optgroup, option, output, p, param, picture, plaintext, pre, progress, q, rb, rp, rt, rtc, ruby, s, samp, script, section, select, shadow, slot, small, source, spacer, span, strike, method, and a second script that used the object property strong, style, sub, summary, sup, svg, table, template, textarea, time, title, track, tt, u, ul, var window.BOOMR.url as the src of a dynamically added TS3 button, fieldset, input, output,

select, textarea script. Attackers can escalate this vulnerable pattern to client- side XSS via double clobbering. First, they clobber the TS4 image, img, object

invoked native method, causing a runtime error when the TS5 a, abbr, acronym, address, applet, area

browser parses the first script. Therefore, the browser stops TS6 basefont, bgsound, blink parsing the rest of the script and BOOMR becomes undefined. TS7 noembed, noframes, noscript, script, style, template, textarea, title, xmp Then, attackers can clobber window.BOOMR.url and TS8 ins, isindex, kbd, keygen, label, legend, li, link, listing, main, map, mark, marquee, consequently control the script src by injecting a DOM menu, menuitem, meta, meter, multicol, nav, nextid, nobr, object, ol, optgroup, option, output, p, param, picture, pre, progress, q, rb, rp, rt, rtc, ruby, s, samp, section, select, Clobbering markup, e.g., . We discovered that TS9 fieldset, figcaption, figure, font, footer, form, h1, header, hgroup, hr, i, image it is possible to inject such non-script markup to the client- side page leveraging the search functionality and the URL TS10 form, iframe, image, img, script, style, table, template

query parameters, which were reflected back to the page. TS11 caption, col, colgroup, tbody, td, tfoot, th, thead, tr

TS12 p, param, picture, plaintext, pre, progress, q, rb, rp, rt, rtc, ruby, s, samp, script, Trello. We discovered that Trello uses a global object section, select, shadow, slot, small, source, spacer, span, strike, strong, style, sub, property called window.ClickTaleScriptSource summary, sup, table, template, textarea, time, title, track, tt, u, ul, var, video, wbr, xmp, a, abbr, acronym, address, applet, area, article, aside, audio, b, base, basefont, bdi, to programmatically load a script named wrScript. bdo, bgsound, big, blink, blockquote, br, button, canvas, center, cite, code, command, However, this property was clobberable as content, data, datalist, dd, del, details, dfn, dialog, dir, div, dl, dt, element, em, embed, fieldset, figcaption, figure, font, footer, form, h1, header, hgroup, hr, i ClickTaleScriptSource was an undefined variable TS13 h1, header, hgroup, hr, i, image, img, input, ins, isindex, kbd, keygen, label, legend, following the vulnerable pattern A of Table 6. Finally, we li, link, listing, main, map, mark, marquee, menu, menuitem, meta, meter, multicol, found that it is possible to insert a persistent, non-script nav, nextid, nobr, noembed, noframes, noscript, object, ol, optgroup, option, output, p, param, picture, plaintext, pre, progress, q, rb, rp, rt, rtc, ruby, s, samp, script, section, markup to overwrite this object property by editing a select, shadow, slot, small, source, spacer, span, strike, strong, style, sub, summary, sup, table, template, textarea, time, title, track, tt, u, ul, var, video, wbr, xmp, a, abbr, comment for a card in Trello boards, which resulted in acronym, address, applet, area, article, aside, audio arbitrary client-side code execution. TS14 form, b, base, basefont, bdi, bdo, bgsound, big, blink, blockquote, br, button, canvas, center, cite, code, command, content, data, datalist, dd, del, details, dfn, dialog, dir, Fandom. We discovered a DOM Clobbering vulner- div, dl, dt, element, em, embed, fieldset, figcaption, figure, font, footer ability in Fandom affecting the users' message wall TS15 data, datalist, dd, del, details, dfn, dialog, dir, div, dl, dt, element, em, embed, fieldset, that resulted in

open redirection. Specifically, the figcaption, figure, font, footer, form, h1, header, hgroup, hr, i, image, img, input, ins, isindex, kbd, keygen, label, legend, li, link, listing, main, map, mark, marquee, menu, JavaScript program contained an assignment to the menuitem, meta, meter, multicol, nav, nextid, nobr, noembed, noframes, noscript, location.href property of the top-level window, whose object, ol, optgroup, option, output, p, param, picture, plaintext, pre, progress, q, rb, rp, rt, b, base, basefont, bdi, bdo, bgsound, big, blink, blockquote, br, button, canvas, value was tainted with a clobberable object property, center, cite, code, command, content, rtc, ruby, s, samp, script, section, select, shadow i.e., form.elements.targetUsername.value. At- TS16 rtc, ruby, s, samp, script, section, select, shadow, slot, small, source, spacer, span, tackers can manipulate the value of this property by, e.g., two strike, strong, style, sub, summary, sup, table, template, textarea, time, title, track, tt, u, ul, var, video, wbr, xmp, a, abbr, acronym, address, applet, area, article, aside, audio, nested iframe tags that are named form and elements, b, base, basefont, bdi, bdo, bgsound, big, blink, blockquote, br, button, canvas, center, and an additional input element in the nested frame. The cite, code, command, content, data, datalist, dd, del, details, dfn, dialog, dir, div, dl, dt, element, em, embed, fieldset, figcaption, figure, font, footer, form, h1, header, hgroup, input is named targetUsername, and has a value hr, i, image containing a malicious URL, which will be set as the window TS17 br, button, canvas, center, cite, code, command, content, data, datalist, dd, del, details, dfn, dialog, dir, div, dl, dt, element, em, embed, fieldset, figcaption, figure, font, footer, URL. We found that it is possible to inject non-script markup form, h1, header, hgroup, hr, i in the page in two distinct ways: (i) attackers can insert TS18 address, applet, area, article, aside, audio, b, base, basefont, bdi, bdo, bgsound, big, persistent payloads using the post functionality in the profile blink, blockquote, br, button, canvas, center, cite, code, command, content, data, datalist, dd, del, details, dfn, dialog, dir message wall, and (ii) a URL parameter in the path was reflected back to the page without extra validation, enabling TS19 form, iframe, script, style, template

transient insertion of clobbering payloads in the page. TS20 image, img, input, noembed, noframes, noscript

A.2. Additional Evaluation Details TABLE 7: List of HTML tags used in Table 2 that share the same DOM Clobbering behaviour.

16 Æ Security Threat ! Semantic Type Description Reference ý JavaScript Sink Client-side Open Redirect WIN LOC WRITE Redirecting the Window URL [36, 37] window.location = T Websocket Hijacking WEBSOCK URL WRITE Hijacking Websocket Connections [80, 81] new WebSocket(T) Cookie Manipulation DOC COOKIE WRITE Manipulating Cookie State [1, 5, 82] document.cookie = T Doc. Domain Manipulation DOC DOMAIN WRITE Bypassing SOP [83, 84] document.domain = T Client-side JSON Injection JSON PARSE Parsing Untrusted JSON [85–87] JSON.parse(T) RegEx Injection REGEX BUILD Injecting Regex for ReDoS [88, 89] new RegExp(T) postMessage Manipulation POST MSG WRITE Manipulating postMessages [82] window.postMessage(T) Local File Path Manipulation FILE PATH WRITE Manipulating Path of Read Files [90] new FileReader().readAsText(T) Cross-site Scripting (XSS) CODE LOADING Loading New Scripts [1, 22, 39] script.src = T CODE EXEC Executing Arbitrary JavaScript [1, 7] script.textContent = T eval(T) setTimeout(T) setInterval(T) new

Function(T) DOM NODE INJECT Injecting DOM Elements [1, 7, 12, 87] document.write(T) document.writeln(T) elm.innerHTML = T elm.outerHTML = T elm.insertAdjacentHTML(T) elm.insertAdjacentElement(T) elm.replaceChild(T) elm.append(T) elm.appendChild(T) Web Storage Manipulation DOC STORAGE WRITE Manipulating Storage State [1, 5, 82] localStorage.setItem() sessionStorage.setItem() Client-side Request Forgery REQ Manipulating Asynchronous Reqs. [32, 133] fetch(T) XMLHttpRequest.open(T) asyncRequest(T) \$.ajax(T) Legend: T= Tainted Variable;

TABLE 8: Summary of primitive JavaScript sinks and semantic types supported by TheThing grouped by the security risk of manipulating the sink object. The list is obtained by aggregating the client-side JavaScript sinks considered in existing literature.

Source ! Semantic Type Graph Queries S1: variable v CLOB CUSTOM VAR Task: Identifying Source Si as in Table 3 S4, S5: window.v CLOB WIN CUSTOM VAR Qs1 = {n: n ∈ N ∧ n.type == 'Identifier' ∧ ∃ v ∈ NP ∧ CLOB(v) ∧ n.name == v } S6: document.v CLOB DOC CUSTOM VAR Qs2 = {n: n ∈ N ∧ ∀ s ∈ N, @ e ∈ E, e == edge(s, n) ∧ e.type == 'PDG' ∧ ∃ v, v ∈ / NP S2: property p CLOB NATIVE PROP ∧ e.value == v ∧ (s.type == 'AssignmentExp' v s.type == 'VarDeclaration') } S3: window.p CLOB WIN NATIVE PROP S7: document.p CLOB DOC NATIVE PROP Qs3 = {n: n ∈ N ∧ n.type == 'MemberExp' ∧ n.object == window ∧ ∃ v ∈ NP ∧ CLOB(v) Legend: Si = case Si in Table 3; ∧ n.property == v }

TABLE 9: Summary of DOM Clobbering sources and their semantic types Qs4 = {n: n ∈ N ∧ n.type == 'MemberExp' ∧ n.object == window ∧ ∃ v ∈ / NP ∧ n.property == v ∧ ∀ s ∈ N, @ e ∈ E, e == edge(s, n) ∧ e.type == 'PDG' ∧ e.value == v ∧ (s.type == based on the seven cases of Table 3. 'AssignmentExp' v s.type == 'VarDeclaration' ∧ s.kind = 'var') }

Qs5 = {n: n ∈ N ∧ n.type == 'MemberExp' ∧ n.object == window ∧ ∃ v ∈ / NP ∧ n.property Sanitizer Version Sanitizer Version == v ∧ ∃ s ∈ N, e ∈ E, e == edge(s, n) ∧ e.type == 'PDG' ∧ e.value == v ∧ (s.type == 'AssignmentExp' v s.type == 'VarDeclaration' ∧ s.kind ∈ {'var', 'let', 'const'})} ∧ ∃ f ∈ Client-side JS Node.js N, f.type == 'CallExp' ∧ f.script == s.script ∧ f.name ∈ NP ∧ CLOB(f.name)

1. DOMPurify 2.3.4 1. Insane 2.6.2
2. Google Closure Lib. 20211201.0.0 2. Bleach 0.3.0 Qs6 = {n: n ∈ N ∧ n.type == 'MemberExp' ∧ n.object == document ∧ ∃ v ∈

/ NP ∧

1. JS-XSS 1.0.10 Bower-angular-sanitize 1.8.2 n.property == v }
2. Sanitize-HTML 2.6.1 Yahoo html-purify 1.1.0
3. Google Caja 6015 Arcgis 2.9.0 Qs7 = {n: n ∈ N ∧ n.type == 'MemberExp' ∧ n.object == document ∧ ∃ v ∈ NP ∧

CLOB(v) ∧ n.property == v } Python PHP

1. Mozilla Bleach 4.1.0 1. Htmlpurifier 4.14.0
2. LXML 4.7.1 2. Html-sanitizer 1.5.0 Task: Identifying Sink Fi as in Table 8
3. HTML Sanitizer 1.9.3 3. Symfony HtmlSanitizer 1.0.0 Qsinks = {n: n ∈ N ∧ ∃ c ∈ N ∧ hasChild(n, c) ∧ c.type == 'Identifier' ∧ c ∈ SI }
4. HtmlLaundry 2.2 4. HTMLawed 1.2

Task: Identifying Vulnerable Sinks C# Java

- TABLE 10: The specific versions of HTML sanitizers tested in §6 TABLE 11: Excerpt of DOM Clobbering detection queries.

95.0.4638

92.0.4515

65.2.3381

82.0.4227

95.0.1020

96.0.1054

95.0.1020 94.1.2

14.7.1

11.0.1

15.0.6

13.3.8 3.2.3 96.0

95.0

39.0

15.1

14.1

13.1 API 75 72 77 70 69 77 75 72 77 75 72 77 76 77
77 77 69 75 75 79 Property 246 244 256 240 238 255 246 244 254 246 244 255 255 258
260 255 244 246 251 268 Total 321 316 333 310 307 332 321 316 331 321 316 322 331
335 337 332 313 321 326 347 Legend: TB= Tor Browser; SI= Samsung Internet; UC= UC
Browser;

TABLE 12: Count of clobbered native DOM APIs in mobile and desktop browsers. Browsers with similar behaviours are grouped with the same color.

. Chrome Firefox Opera Edge Safari TB SI UC Chrome Firefox Opera Edge Safari TB SI UC
95.0.4638

15.1 14.1 13.1 API `cancelIdleCallback()`
`sidebar clearImmediate()` `Notification convertPointFromNodeToPage()` `dataLayer`
`convertPointFromPageToNode()` `XDomainRequest createImageBitmap()` `ActiveXObject`
`dump()` `attachEvent` `getDefaultComputedStyle()` `EventEmitter` `home()` `ReportingObserver`
`minimize()` `chrome` `openDialog()` `safari` `print()` `firefoxAccess` `requestIdleCallback()`
`trustedTypes` `routeEvent()` `NaN` `scrollByLines()` `SharedArrayBuffer` `scrollByPages()` `afterprint`
`setCursor()` `animationcancel` `setImmediate()` `animationend` `showDirectoryPicker()`
`animationiteration` `showModalDialog()` `beforeprint` `showOpenFilePicker()` `beforeunload`
`showSaveFilePicker()` `copy` `sizeToContent()` `cut` `updateCommands()` `DOMContentLoaded`
`error` `caches` `hashchange` `controllers` `languagechange` `crossOriginIsolated` `load`
`dialogArguments` `message` `directories` `messageerror` `fullScreen` `offline`
`mozAnimationStartTime` `online` `mozInnerScreenX` `orientationchange` `mozInnerScreenY`
`pagehide` `pageshow` `onappinstalled` `paste` `onauxclick` `popstate` `onbeforeinstallprompt`
`rejectionhandled` `oncancel` `Storage` `onclose` `transitioncancel` `ondeviceorientationabsolute`
`unhandledrejection` `ondragdrop` `unload` `onformdata` `vrdisplayconnect`
`ongamepadconnected` `vrdisplaydisconnect` `ongamepaddisconnected`
`vrdisplaypresentchange` `onloadend` `onanimationcancel` `onmessageerror` `ontouchcancel`
`onpaint` `ontouchstart` `onvrdisplayactivate` `defaultStatus` `onvrdisplayblur` `Touch`
`onvrdisplayconnect` `TouchEvent` `onvrdisplaydeactivate` `ondevicemotion`
`onvrdisplaydisconnect` `ondeviceorientation` `onvrdisplayfocus` `ontransitioncancel`
`onvrdisplaypointerrestricted` `speechSynthesis` `onvrdisplaypointerunrestricted`
`onselectionchange` `onvrdisplaypresentchange` `MediaSource` `pkcs11` `onselectstart`

scrollMaxX scrollMaxY Legend: TB= Tor Browser; SI= Samsung Int.; UC= UC Browser; =
successfully clobbered; = clobbering fails;

TABLE 13: List of clobbered Window methods and properties in web browsers.

18